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⁶**Game-Theory-Based Balance: Formalizing the Machine Unlearning Problem as a Game
Between Model Provider and Data Provider**

PhD Proposal Submitted to



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1. Introduction

⁸ The rapid advancements in Artificial Intelligence (AI) and Machine Learning (ML) have led to widespread adoption across various applications, significantly improving fields ⁵ such as computer vision, speech recognition, and medical diagnosis. However, this data-intensive progress comes with increasing concerns regarding **user privacy**.

³ As the right to be forgotten has been legislated worldwide, ³ Notable instances include the GDPR in the European Union [16], the PIPEDA privacy legislation in Canada [17], and the California Consumer Privacy Act in the United States [18] ¹² and Digital Personal Data Protection Act, 2023 in India ⁷ [19].

According to these laws, companies must take reasonable measures to guarantee that personal data is deleted upon request. ³ Numerous studies have sought to design unlearning methods to protect users' privacy when they want to remove their data from machine learning service platforms. This has given rise to the concept of "machine unlearning" — a field dedicated to removing the influence of specified samples from trained models.

Machine unlearning is a vital capability in modern AI systems, allowing models to selectively forget specific data to uphold privacy rights, correct biases, and comply with regulations like GDPR. Yet, the process is not merely technical—it involves strategic decisions between the model provider, who must weigh utility loss and compliance cost, and the data provider, who demands control over their personal data.

Game theory has emerged as a powerful tool in privacy-preserving techniques over the past decade, offering structured models to navigate ¹⁰ the trade-off between data privacy and utility. Numerous schemes have leveraged game-theoretic principles to model interactions among stakeholders, such as hybrid encrypted cloud data deduplication and personalized differential privacy. In the context of machine unlearning, this intersection is particularly pertinent. For a

model provider, unlearning inherently presents a trade-off: an over-unlearning strategy may degrade model performance, while insufficient protection risks privacy leaks.

This research aims to formalize the machine unlearning problem as a strategic game between two key players—the model provider and the data provider. By defining their strategies and utilities within a game-theoretic framework, we aim to identify equilibrium behaviours that balance performance preservation with privacy guarantees. This approach transforms unlearning from a reactive mechanism into a principled, auditable process, offering a logically cohesive solution to a complex and increasingly critical challenge in AI.

The field of Machine Unlearning is an expansive and rapidly evolving domain. Based on a comprehensive review of the current literature and trends, There are several key research areas with significant potential for innovation. While the field is vast, specific topics have emerged as particularly promising due to their broad applicability and the critical challenges they address. These areas of interest include:

- **Federated Unlearning:** This domain offers numerous research avenues, including its application in federated settings, the integration of game theory, the development of theoretical frameworks, and the handling of dynamic, real-world unlearning requests.
- **LLM Unlearning:** As a topic of significant commercial and academic interest, unlearning for large language models (LLMs) represents a crucial use case. I have incorporated this as a core application within the theoretical framework of my research.
- **Feature Unlearning:** Given the established work in feature extraction, selection, and generation, this topic presents a compelling opportunity to explore a novel dimension of unlearning by shifting the focus from individual data samples to their constituent features.

- **Game Theory in Machine Unlearning:** This is a highly compelling topic with wide-ranging applicability. Game theory can be leveraged in diverse scenarios, from federated and LLM-based unlearning to designing incentive mechanisms and developing both algorithmic-specific and algorithmic-agnostic approaches.
- **Theoretical Frameworks:** Initially considered too generic, the development of a formal theoretical framework is now recognized as a vital research objective. While this might be a complex undertaking, it provides the flexibility to establish a rigorous foundation for any chosen research topic, ensuring formal guarantees of effectiveness. The concepts of unlearning evaluation and verification are naturally subsumed within this framework.
- **Sequential Unlearning and Recommender Systems:** These areas represent a distinct dimension of unlearning, particularly in dynamic environments. They offer fertile ground for future research, including applications in federated learning, sequential and session-based recommendations, and LLMs.

A fundamental challenge in machine unlearning is the inherent **trade-off between preserving user privacy and maintaining model performance and usability**. Overly aggressive unlearning strategies can lead to significant degradation in model accuracy and robustness, rendering the model less useful in practice. Conversely, insufficient protection risks privacy leaks, as sensitive information about unlearned data may still be inferable from the model. This proposal posits that **game theory**, a powerful analytical framework for studying strategic interactions and competitions among all system stakeholders, can provide a robust mechanism to formalize and navigate this complex balance.

2. Literature Review

- [1] Machine Unlearning: A Survey HENG XU, TIANQING ZHU, and LEFENG ZHANG, University of Technology Sydney, Australia WANLEI ZHOU, City University of Macau, China PHILIP S. YU, University of Illinois at Chicago, United States
- [2] Machine Unlearning: A Comprehensive Survey WEIQI WANG, University of Technology Sydney, Australia ZHIYI TIAN, University of Technology Sydney, Australia CHENHAN ZHANG, University of Technology Sydney, Australia SHUI YU, University of Technology Sydney, Australia
- [3] Stefanos Leonardos and Georgios Piliouras. 2022. Exploration-exploitation in multi-agent learning: Catastrophe theory meets game theory. *Artif. Intell.* 304 (2022), 103653. DOI:<http://dx.doi.org/10.1016/j.artint.2021.103653>
- [4] Lei Cui, Youyang Qu, Mohammad Reza Nosouhi, Shui Yu, Jianwei Niu, and Gang Xie. 2019. Improving data utility through game theory in personalized differential privacy. *J. Comput. Sci. Technol.* 34, 2 (2019), 272–286. DOI:<http://dx.doi.org/10.1007/s11390-019-1910-3>
- [5] Xueqin Liang, Zheng Yan, Robert H. Deng, and Qinghua Zheng. 2021. Investigating the adoption of hybrid encrypted cloud data deduplication with game theory. *IEEE Trans. Parallel Distrib. Syst.* 32, 3 (2021), 587–600. DOI:<http://dx.doi.org/10.1109/TPDS.2020.3028685>
- [6] Jiahui Geng, Qing Li, Herbert Woisetschlaeger, Zongxiong Chen, Fengyu Cai, Yuxia Wang, Preslav Nakov, Hans-Arno Jacobsen, Fakhri Karray A Comprehensive Survey of Machine Unlearning Techniques for Large Language Models DOI: <https://doi.org/10.48550/arXiv.2503.01854>
- [7] Exploring Federated Unlearning: Review, Comparison, and Insights Yang Zhao, Member, IEEE, Jiaxi Yang, Student Member, IEEE, Yiling Tao, Student Member, IEEE,

Lixu Wang, Student Member, IEEE, Xiaoxiao Li, Member, IEEE, Dusit Niyato, Fellow, IEEE, and H. Vincent Poor, Life Fellow, IEEE

[8] Chuan Guo, Tom Goldstein, Awni Y. Hannun, and Laurens van der Maaten. 2020. Certified data removal from machine learning models. In 37th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 119). PMLR, 3832–3842. Retrieved from <http://proceedings.mlr.press/v119/guo20c.html>.

[9] Reinforcement Unlearning Dayong Ye*, Tianqing Zhu†, Congcong Zhu†, Derui Wang‡, Kun Gao*, Zewei Shi‡, Sheng Shen, Wanlei Zhou†, and Minhui Xue‡
*University of Technology Sydney, †City University of Macau, ‡CSIRO's Data61, Torrens University Australia, <https://arxiv.org/pdf/2312.15910>

[10] Xiaofeng Ding, Hongbiao Fang, Zhilin Zhang, Kim-Kwang Raymond Choo, and Hai Jin. 2022. Privacy-preserving feature extraction via adversarial training. IEEE Trans. Knowl. Data Eng. 34, 4 (2022), 1967–1979. DOI:<http://dx.doi.org/10.1109/TKDE.2020.2997604>

[11] Bo Zhang, Hancheng Ye, Gang Yu, Bin Wang, Yike Wu, Jiayuan Fan, and Tao Chen. 2022. Sample-centric feature generation for semi-supervised few-shot learning. IEEE Trans. Image Process. 31 (2022), 2309–2320. DOI:<http://dx.doi.org/10.1109/TIP.2022.3154938>

[12] Jiajing Zhu, Yongguo Liu, Chuanbiao Wen, and Xindong Wu. 2022. DGDFS: Dependence Guided Discriminative Feature Selection for predicting adverse drug-drug interaction. IEEE Trans. Knowl. Data Eng. 34, 1 (2022), 271–285. DOI:<http://dx.doi.org/10.1109/TKDE.2020.2978055>

[13] [Wu et al., 2023] Xinwei Wu, Junzhuo Li, Minghui Xu, Weilong Dong, Shuangzhi Wu, Chao Bian, and Deyi Xiong. DEPN: Detecting and editing privacy neurons in

pretrained language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 2875–2886, Singapore, December 2023. Association for Computational Linguistics.

[14] [Jia et al., 2024] Jinghan Jia, Jiancheng Liu, Yihua Zhang, Parikshit Ram, Nathalie Baracaldo, and Sijia Liu. WAGLE: Strategic weight attribution for effective and modular unlearning in large language models. In The Thirty-eighth Annual Conf

[15] A Survey on Recommendation Unlearning: Fundamentals, Taxonomy, Evaluation, and Open Questions Yuyuan Li, Xiaohua Feng, Chaochao Chen*, Senior Member, IEEE, and Qiang Yang, Fellow, IEEE

[16] Alessandro Mantelero. The EU proposal for a general data protection regulation and the roots of the ‘right to be forgotten’. Comput. Law Secur. Rev., 29(3):229–235, 2013.

[17] O. of the Privacy Commissioner of Canada. Announcement: Privacy commissioner seeks federal court determination on key issue for Canadians’ online reputation.

[18] Lydia de la Torre. A guide to the California Consumer Privacy Act of 2018. Available at SSRN 3275571, 2018.

[19] Digital Personal Data Protection Act, 2023. Available at <https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf>

[TBD] #Add Research Papers Here

3. Research Gaps identified

Based on this exploration, I have listed few core research areas, each addressing a unique and critical aspect of machine unlearning:

- **Game-Theory-Based Balance:** Formalizing the Machine Learning Problem as a Game Between Model Provider and Data Provider.
- **Feature-Based Unlearning Methods:** Shifting Influence Calculation from Samples to Features.
- **Developing a Comprehensive Theoretical Framework for LLM Unlearning.**
- **Advancing Recommendation Unlearning in Dynamic and User-Centric Scenarios.**
- **Towards a Unified and Certifiable Framework for Reinforcement Unlearning in Dynamic and Safety-Critical Environments.**

Game-theory-based balance [1]: Game theory has been a booming field with several representative privacy-preserving techniques coming out in the past decade [3]. There are many schemes involving privacy-preserving solutions based on game theory that trade off data privacy and utility issues [4, 5]. For a model provider, machine unlearning is also a tradeoff between model performance and user privacy, where an over-unlearning strategy may lead to performance degradation, while insufficient protection may lead to privacy leaks. Can we formalize the unlearning problem as a game between two players: a model provider and a data provider? If so, then we could provide a game model between these two entities and determine a set of strategies and utilities to figure out how to perform unlearning operations that maintain the model's performance to the maximum extent possible. Such an approach could also protect the user's sensitive data from being leaked. These are open issues that need to be explored further.

Incentive Mechanism Design [7]. Incentive mechanisms that balance the interest of users and system efficiency are crucial in motivating participation in federated unlearning. Users should be able to efficiently ask for their data to be removed without any substantial loss of the

accuracy of the model. Designing such mechanisms is an attractive way for possible future work in this area.

Federated Unlearning for Large Language Models (LLMs) [7]. LLMs are distinguished by their extensive scale, containing billions to trillions of parameters. For unlearning, LLMs do not store data in a straightforward way that allows for easy removal. Data in these models are distributed across many parameters. Thus, a challenge for future work is to discover methods to unlearn data in the LLM setting.

4. Objectives

Primary Objective

To formalize the machine unlearning problem as a game between a model provider and a data provider, establishing a theoretical framework to identify optimal strategies that balance model performance/utility with user privacy protection.

Secondary Objectives

- To **develop a comprehensive game model** (e.g., a Stackelberg or Static Bayesian game) that captures the strategic interactions and interdependent utilities of a model provider and a data provider in the context of machine unlearning.
- To **define precise utility functions** for both entities: for the model provider, considering factors like model accuracy, computational cost of unlearning, and robustness; and for the data provider, focusing on privacy guarantees (e.g., measured by information leakage prevention) and retained data value.
- To **identify and analyse equilibrium strategies** within the proposed game model that facilitate unlearning operations, aiming to achieve an optimal trade-off point where the collective utility of both parties is maximized.

- To **propose an algorithmic mechanism** to compute and implement these game-theoretic optimal unlearning strategies, demonstrating its feasibility and convergence to the derived equilibria.
- To **evaluate the proposed framework** using established metrics for model performance (e.g., accuracy, consistency) and privacy protection (e.g., membership inference attack success rates, information bounds), providing empirical evidence of its effectiveness.

5. Detailed Methodology

Research Design This research will primarily involve theoretical modelling and simulation-based experimentation.

- **Literature Review:** A critical literature review on the application of game theory in two distinct areas of computer science. First, an in-depth analysis of ¹⁰ Lei Cui et al.'s "Improving data utility through game theory in personalized differential privacy" will be conducted. This will involve a detailed examination of the **Static Bayesian Game Theory Approach** applied to recommender systems, with a specific focus on the underlying mathematical principles, including ¹⁵ the concept of Nash equilibrium. The review will then explore the potential for applying alternative game theory models to new scenarios.

Second, the paper by Xueqin Liang et al. on "Investigating the adoption of hybrid encrypted cloud data deduplication with game theory" will be analyzed. The focus here will be on the **Stackelberg game model** used for data deduplication. The analysis will extend to evaluate how this specific game theory approach can be adapted and applied within the context of **Machine Unlearning** algorithms.

- **Game Model Selection and Development:** We will investigate and select the most appropriate game theory model.
 - **Evaluate the suitability of different Game Theory models** for (Classifications of Game Theory Models) various machine unlearning applications. The analysis should identify which specific models are most effective for optimizing the unlearning process to achieve an efficient and desirable outcome.
 1. How can classifications of Game Theory models be applied to the problem of machine unlearning? Specifically, which models (e.g., cooperative, non-cooperative, static, dynamic) are most appropriate for different unlearning scenarios, and what are the implications for achieving an efficient and effective outcome?
 2. A **Stackelberg game** [5] could be suitable if there's a clear leader-follower dynamic, e.g., the model provider sets an unlearning policy, and data providers react by deciding their unlearning requests. Backward induction would be used ¹⁴ to find the sub-game perfect Nash Equilibrium.
 3. Alternatively, a **Static Bayesian game** [4] could model scenarios with incomplete information, where each player is uncertain about the other's types or payoff functions (e.g., the model provider doesn't know a user's exact privacy sensitivity, or a user doesn't know the model provider's true unlearning cost). This aligns with insights from personalized differential privacy.

- **Player and Strategy Definition:**

- **Players:** The Model Provider (MP) and the Data Provider (DP).
- **MP Strategies:** Varying the **intensity or type of unlearning** (e.g., exact vs. approximate, specific unlearning algorithms), the **frequency of unlearning operations**, or the **pricing/incentive structure** for unlearning services.
- **DP Strategies:** Deciding **whether and when to request unlearning** for specific data, potentially based on perceived privacy risks or model utility.

- **Utility Function Formulation:**

- **MP Utility:** Will incorporate **model performance** (e.g., accuracy, robustness post-unlearning), **computational cost of unlearning** (time, resources), and potential **reputational gains/losses** associated with privacy compliance.
- **DP Utility:** Will incorporate the **level of privacy protection achieved** (quantified by reductions in privacy leakage attacks like membership inference or reconstruction attacks), the **value of retained data** (if partial unlearning), and the **convenience/cost of initiating unlearning requests**.

- **Equilibrium Analysis:** We will analytically derive the Nash Equilibrium (or Stackelberg Equilibrium) of the formalized game. This will involve solving optimization problems for each player's utility function, given the other players' strategies. We will investigate the conditions under which a unique or multiple equilibria exist.

- **Integration of Catastrophe Theory:** By framing the "exploration" (unlearning) parameter as a control variable, we will analyse how changes in unlearning intensity or frequency can lead to **phase transitions** in the game's equilibrium landscape,

potentially resulting in abrupt shifts in model performance or privacy outcomes. This will help identify "provably beneficial or damaging" unlearning scenarios. The smooth Q-learning dynamics, which converge to QRE and are linked to catastrophe theory, will be considered for insights into agent behavior with bounded rationality.

6. Expected outcome

- A **formalized game model**, potentially a multi-stage Stackelberg or static Bayesian game, describing the strategic interactions between a model provider and a data provider in machine unlearning.
- **Derived equilibrium strategies** that illustrate optimal trade-offs between model accuracy and privacy protection, considering varying unlearning costs and privacy sensitivities. This would identify the **optimal balance** point where the model's performance is minimally impacted while achieving maximum feasible privacy for the unlearned data.
- **Demonstration of phase transitions:** Through simulation and analysis, we expect to show how gradual changes in unlearning parameters (e.g., "exploration rates" in the context of SQL dynamics) can lead to abrupt shifts in equilibrium states, impacting overall system performance (e.g., identifying scenarios where unlearning is provably beneficial or damaging).
- A **gradient-based algorithm** (or similar iterative method) that helps both the model provider and data provider converge to their near-optimal strategies, providing a practical tool for implementing the game-theoretic balance.
- Quantifiable results showing the effectiveness of the game-theoretic approach in **reducing privacy leakage** (e.g., lower membership inference attack success rates after unlearning) while **maintaining competitive model accuracy**.

Theoretical Contributions

- **A Novel Game-Theoretic Framework for Machine Unlearning:** This research will introduce the first formal game-theoretic model specifically designed to capture the strategic interactions between a model provider and a data provider in machine unlearning. This will bridge a significant ¹³gap in the current literature by providing a principled approach to the performance-privacy trade-off.
- **Formalization of Strategic Interactions:** We will provide precise definitions of strategies and utility functions for both model providers and data providers in the unlearning context, accounting for their often conflicting objectives (e.g., model utility vs. privacy demands).
- **Optimal Equilibrium Identification:** By deriving Nash or Stackelberg Equilibria, the research will identify optimal, stable strategies for performing unlearning operations, offering guidelines on how to maximize collective utility while respecting individual incentives.
- **Insights from Catastrophe Theory:** Integrating concepts from catastrophe theory will provide novel insights into the non-linear effects of unlearning parameters on model performance and privacy, explaining phenomena like abrupt phase transitions and path-dependence in learning algorithms.

Practical Implications

- **Guidelines for Ethical AI Development:** The framework will provide actionable insights for developers and organizations to design and implement machine unlearning algorithms that are not only efficient but also ethically compliant with privacy regulations like GDPR, balancing model integrity with user rights.

- **Optimized Unlearning Policies:** Model providers will gain a clearer understanding of how to set their unlearning policies (e.g., unlearning frequency, intensity, and associated costs/incentives) to maintain high model quality while effectively responding to user deletion requests.
- **Empowering Data Providers:** Data providers (users) will be empowered with a better understanding of the dynamics involved in unlearning, enabling them to make more informed decisions about their data and privacy requests, and to assess the effectiveness of unlearning services.
- **Benchmarking and Verification:** The proposed analytical tools and evaluation metrics will serve as a robust benchmark for future unlearning research and provide a more comprehensive approach to verifying the effectiveness of unlearning processes beyond current empirical methods.

7. Importance of proposed research investigation and mention the link to relevant SDG

The proposed research contributes to SDG 9 (Industry, Innovation, and Infrastructure) by advancing responsible AI methods that enhance digital trust and innovation. It also supports SDG 16 (Peace, Justice, and Strong Institutions) by ensuring privacy-preserving mechanisms that uphold fairness and accountability in AI systems. Additionally, the work aligns with SDG 17 (Partnerships for the Goals) through its potential to foster interdisciplinary collaboration in responsible AI research. Secondary contributions include SDG 4 (Quality Education) and SDG 12 (Responsible Consumption and Production), reflecting the educational and sustainability dimensions of the study.

 Relevant SDGs for Your Research

SDG	Title	Connection to Your Research	Link
SDG 9	Industry, Innovation, and Infrastructure	Advances responsible AI methods that strengthen digital infrastructure and foster innovation.	SDG 9
SDG 16	Peace, Justice, and Strong Institutions	Privacy-preserving AI supports justice, trust, and accountability in institutions.	SDG 16
SDG 17	Partnerships for the Goals	Enables interdisciplinary collaboration across academia, industry, and policy for responsible AI.	SDG 17
SDG 4	Quality Education	Integrates machine unlearning into AI curricula, contributing to ethical and accessible education.	SDG 4
SDG 12	Responsible Consumption and Production	Promotes sustainable digital practices by reducing unnecessary retention of personal data.	SDG 12

8. Research Time plan

Research Activity (TEMPLATE-to be edited as per the requirement)	Time in Months (from Registration)									
	06	12	18	24	30	36	42	48	54	
Course Work										
Literature Survey										
Methodology Step 1										
Methodology Step 2										
Methodology Step 3										
Methodology Step 4										
DAC presentation and registration renewal										

Conference presentations				C		C			
Journal Publications					J		J		
Preparation of Final Thesis									
Synopsis presentation									↑
Submission of Thesis									↑

Part-time candidates - 4 years 6 months (54 months)

9. Pilot study / Preliminary work done

9. Pilot Study / Preliminary Work Done

The pilot study was undertaken to validate the feasibility of combining **Machine Unlearning** techniques with **Game-Theoretic frameworks**. Since direct implementations of game theory in unlearning are scarce, the initial groundwork focused on:

- Implementing existing **membership inference** and **machine unlearning** papers to understand their methodologies.
- Exploring datasets and experimental setups to benchmark performance.
- Identifying gaps in current approaches that could be addressed through game-theoretic modelling (e.g., incentive compatibility, fairness in unlearning requests).

This preliminary work provides a foundation for the proposed methodology, where **Game Theory will be applied to model strategic interactions between data owners and service providers in unlearning scenarios.**

◆ **Papers Implemented**

S. No.	Title of Paper	Authors / Year	Key Methodology Implemented	Tools / Frameworks Used	Key Findings / Observations	Relevance to Proposed Research
1	<i>Membership Inference via Backdoor-ing (MIB)</i>	Hu et al., 2022	Membership inference using backdoor triggers and hypothesis testing	Python, PyTorch; CIFAR-10, Purchase-100, Location-30 datasets	Demonstrated that marking only 0.1% of data enables effective membership inference with statistical guarantees; ASR thresholds validated	Provides a detection mechanism for unauthorized data usage, an essential precursor to unlearning

S. No.	Title of Paper	Authors / Year	Key Methodology Implemented	Tools / Frameworks Used	Key Findings / Observations	Relevance to Proposed Research
2	<i>Machine Unlearning (SISA Training)</i>	Bourtoul et al., 2020	Sharded, Isolated, Sliced, Aggregated training for efficient unlearning	Python, TensorFlow; Purchase dataset, SVHN, ImageNet	Achieved $4.63\times$ speed-up in unlearning compared to retraining from scratch, with minimal accuracy degradation	Establishes a scalable baseline for unlearning, which can be extended with game-theoretic strategies

◆ Datasets / Experimental Setup

• Datasets Used:

- **CIFAR-10** (image classification, 60,000 samples, 10 classes).
- **Purchase-100** (binary dataset, 197,324 samples, 100 classes).
- **Location-30** (binary dataset, 5,010 samples, 30 classes).
- **SVHN** (digit recognition, used in SISA experiments).
- **ImageNet** (large-scale image classification, used in SISA experiments).

- **Preprocessing Steps:**
 - Normalization of image datasets (CIFAR-10, SVHN, ImageNet).
 - Feature encoding for binary datasets (Purchase-100, Location-30).
 - Trigger embedding for MIB experiments (imperceptible backdoor patterns).
 - **Evaluation Metrics:**
 - **Attack Success Rate (ASR)** for membership inference.
 - **Accuracy degradation** post-unlearning.
 - **Speed-up factor** compared to retraining from scratch.
 - **Statistical significance (t-test)** for hypothesis validation.
 - **Computational Environment:**
 - Experiments conducted on GPU-enabled systems (NVIDIA RTX series).
 - Frameworks: PyTorch (for MIB), TensorFlow (for SISA).
 - Cloud setup tested for scalability (Google Colab / GCP).
-

◆ Key Insights

- **Membership Inference via Backdooring (MIB):**
 - Effective detection of unauthorized data usage with minimal marking ratio (0.1%).
 - Black-box access suffices, making it practical for MLaaS scenarios.
 - Limitation: Focused on detection, not on subsequent unlearning.
- **SISA Training:**
 - Provides computationally feasible unlearning with provable guarantees.
 - Speed-up achieved without significant accuracy loss for simple tasks.

- Limitation: Lacks incentive structures and fairness modeling — potential area for game-theoretic integration.
- **Gap Identified:**
 - Current unlearning methods focus on efficiency but ignore **strategic interactions** (e.g., balancing user privacy demands vs. provider resource costs).
 - Game Theory can formalize these negotiations, ensuring fairness and incentive compatibility.

◆ **Conclusion**

The pilot study confirms the **feasibility of integrating Game Theory into Machine Unlearning**.

- MIB establishes a **detection pipeline** for unauthorized data usage.
- SISA provides a **baseline unlearning framework** with efficiency gains.
- Together, these implementations highlight the need for a **game-theoretic layer** that models unlearning requests as strategic moves, balancing privacy guarantees with computational costs.

This preliminary work provides a strong foundation for the detailed methodology in the proposed research, ensuring that subsequent experiments are grounded in validated implementations.

-
10. Details of expenses and source of funding (detailed break-up of research related expenses for consumables/software/equipment/travel etc. to be given along with details of source of funding. If the study does not require funding, same to be mentioned with justification. Scholarships/fellowships awarded to the

candidate should NOT be mentioned under research related expenses mentioned above.

No	Item	Year 1	Year 2	Year 3	Total
	Consumables 1 ...				
	Consumables 2 ...				
	Travel				
	Total Budget				

11. References: Vancouver or Harvard or APA style etc., as recommended by DAC.

Referencing should be uniform throughout the protocol.

[TBD] #Take Input

12. Course work details suggested by DAC with credits

[TBD] #Take Input

Serial No.	Course name	Credits
1.	Research Methodology	4
2.	Research and Publication Ethics	2
3.	Course 3...	
4.	Course 4...	

13. Similarity check report having *total word count, index < 15% and exclude*

Matches<3 duly signed by the candidate and guide.

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ORIGINALITY REPORT

25%

SIMILARITY INDEX

17%

INTERNET SOURCES

13%

PUBLICATIONS

13%

STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Manipal Academy of Higher Education (MAHE) Student Paper	8%
2	Heng Xu, Tianqing Zhu, Lefeng Zhang, Wanlei Zhou, Philip S. Yu. "Machine Unlearning: A Survey", ACM Computing Surveys, 2023 Publication	5%
3	arxiv.org Internet Source	4%
4	Submitted to Blackburn College, Lancashire Student Paper	3%
5	export.arxiv.org Internet Source	2%
6	Submitted to RMIT University Student Paper	1%
7	Wang, Weiqi. "A Study on Machine Unlearning", University of Technology Sydney (Australia), 2025 Publication	1%
8	publication-cpas-egypt.com Internet Source	1%
9	Submitted to GMR Institute of Technology Student Paper	<1%
10	Lei Cui, Youyang Qu, Mohammad Reza Nosouhi, Shui Yu, Jian-Wei Niu, Gang Xie. "Improving Data Utility Through Game Theory	<1%

11

Jimcymol James, Anjan Gudigar, U. Raghavendra, Jyothi Samanth et al. "Automated Identification of Left Ventricular Hypertrophy using Cardiac Ultrasound imaging: A Systematic Review of Artificial Intelligence Driven Approaches", Informatics in Medicine Unlocked, 2025

Publication

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Publication

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Komal Singh Gill, Anju Sharma, Sharad Saxena. "A Systematic Review on Game-Theoretic Models and Different Types of Security Requirements in Cloud Environment: Challenges and Opportunities", Archives of Computational Methods in Engineering, 2024

Publication

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